Fiscal Year:	FY 2017	**Task Last Updated:**	FY 09/19/2016
PI Name:	Hyers, Robert Ph.D.		
Project Title:	Unified Support for THERMOLAB - ISS, ICOPROSOL, and PARSEC (NNX16AB40G)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Worcester	**State:**	MA
Zip Code:	01609-2280	**Congressional District:**	2
Comments:	Formerly with University of Massachusetts at Amherst.		
Project Type:	Flight	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	11/16/2015	**End Date:**	11/15/2020
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
Contact Monitor:	Craven, Paul	**Contact Phone:**	256-544-7649
Contact Email:	pcravenRRR@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lee, Jonghyun Ph.D. (UMass Amherst)		
Grant/Contract No.:	NNX16AB40G		
Performance Goal No.:			
Performance Goal Text:			
	NOTE: Continuation of "Unified Support for THERMOLAB - ISS, ICOPROSOL, and PARSEC," grant NNX10AR71G with the same investigator, Dr. Robert Hyers. Robert Hyers is U.S. Team Member on ESA-sponsored projects, "Peritectic Alloy Rapid Solidification with Electromagnetic Convection (PARSEC)," "Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS)," and "Thermophysical properties and solidification behaviour of undercooled Ti-Zr-Ni liquids showing an icosahedral short-range order (ICOPROSOL)." Funding is for Dr. Hyers's role as Team Member on the subprojects described below, associated with European Space		

Task Description:

Agency (ESA)-sponsored main projects as noted.

Main project: Peritectic Alloy Rapid Solidification with Electromagnetic Convection (PARSEC); Science Coordinator is Dr. Thomas Volkmann, German Aerospace Center DLR, Institut für Materialphysik im Weltraum, Linder Höhe, Köln, Germany

Subprojects:

1. Transformation Kinetics-ISS (International Space Station). Task description: Using results from thermophysical property evaluation and combining these with the Magnetohydrodynamic (MHD) convection definition, we will set convection over as wide a range as is possible and conduct recalescence studies from the undercooled melt using the Material Science Laboratory Electromagnetic Levitator (MSL-EML) to compare to results from ground-based EML and ESL. The critical parameters of interest are evaluation of the incubation time between formation of metastable phases and subsequent transformation to the stable phase for Fe-based and titanium aluminide alloys. Since this transformation is very rapid, the high-speed video system must be used to image the processes associated with nucleation and growth during phase selection.

2. Thermophysical Properties. Task description: The density, viscosity, and surface tension of the flight samples will be measured in flight for a range of temperatures. Density will be determined by the video metrology, and surface tension and viscosity by the oscillating drop method. This WP includes determination of parameter modifications required for these measurements as well as collection, reduction, and analysis of the resulting data. The resulting data will support the whole range of theoretical and modeling efforts in this proposal.

3. Modeling of Macroconvection. Task description: Magnetohydrodynamic (MHD) model of macroconvection in the samples for different combinations of experimental parameters. Evaluation of fluid velocity, shear rates, turbulence intensity, and their distribution for control of experiments and input to other theorerical and modeling efforts. Collaboration on development of parameters for other work packages to enable the use of convective state as an experimental parameter.

Main project: Thermophysical properties and solidification behaviour of undercooled Ti-Zr-Ni liquids showing an icosahedral short-range order (ICOPROSOL); Science Coordinator is Dr. Dirk Holland-Moritz, Deutsches Zentrum für Luft- und Raumfahrt (DLR), Institut für Materialphysik im Weltraum, Linder Höhe, Köln, Germany.

Subproject:

Modeling of macroconvection: Magnetohydrodynamic modelling of macroconvection in Ti-Zr-Ni samples for different combinations of experimental parameters. Evaluation of fluid velocity, shear rates, turbulence intensity, and their distribution for control of experiments and input to other theorerical and modeling efforts.

Main project: Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS); Science Coordinator is Professor Hans-Joerg Fecht, Ulm University, Ulm, Germany

Subprojects:

Long Duration Microgravity Experiments - ISS. Evaluation of microgravity experiments for fluid flow investigations.

MODELLING A: Thermodynamics, Phase Selection. Modelling of nucleation under various flow conditions for steel alloys.

MODELLING B: Non-contact thermophysical. Modeling of magnetohydrodynamic convection and shear in 1-g, microgravity for 4-6 alloys. Modeling of Marangoni convection in 1-g ESL drops for comparison with microgravity experiments. Modeling of temperature distribution in 1-g and microgravity samples, 4-6 alloys. Input into design of experimental parameters for 4-6 Alloys.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

This project consists of modeling in support of three different flight experiments:
- PARSEC - Peritectic Alloy Rapid Solidification with EM Convection

- ThermoLAB - Thermophysical Properties of Liquid Metallic Alloys

- ICOPROSOL - Thermophysical properties and solidification behaviour of undercooled Ti-Zr-Ni liquids showing an icosahedral short-range order

In PARSEC, we are investigating the effect of fluid flow on the solidification path of certain alloys. Control of the solidification path would enable tailoring of the microstructure and properties of metal parts for specific applications.

In ThermoLAB, we are investigating the thermophysical properties of high-temperature materials. Many of these materials are used commercially. A better understanding of the physical properties will allow more efficient and more reliable production of metallic parts using these alloys. Other alloys are chosen because of scientific interest in their properties, which may illuminate fundamental physics related to the formation of bulk metallic glasses and quasicrystals.

In ICOPROSOL, we are investigating the nucleation and growth of quasicrystals, and the effect of atomic-scale order on the macroscopic properties of these alloys. This fundamental investigation may improve our ability to tailor the microstructure of metals for commercial applications.

Task Progress:	This project consists of modeling in support of three different flight experiments: - PARSEC - Peritectic Alloy Rapid Solidification with EM Convection - ThermoLAB - Thermophysical Properties of Liquid Metallic Alloys - ICOPROSOL - Thermophysical properties and solidification behavior of undercooled Ti-Zr-Ni liquids showing an icosahedral short-range order In PARSEC, we are investigating the effect of fluid flow on the solidification path of certain alloys. Control of the solidification path would enable tailoring of the microstructure and properties of metal parts for specific applications. In ThermoLAB, we are investigating the thermophysical properties of high-temperature materials. Many of these materials are used commercially. A better understanding of the physical properties will allow more efficient and more reliable production of metallic parts using these alloys. Other alloys are chosen because of scientific interest in their properties, which may illuminate fundamental physics related to the formation of bulk metallic glasses and quasicrystals. In ICOPROSOL, we are investigating the nucleation and growth of quasicrystals, and the effect of atomic-scale order on the macroscopic properties of these alloys. This fundamental investigation may improve our ability to tailor the microstructure of metals for commercial applications. Highlights of Work Accomplished During Grant Period 2016-17. • Supported the magnetohydrodynamic simulations for the prediction of phase selection during solidification for PARSEC. • Measured thermophysical properties of Zr-O using ESL at NASA Marshall Space Flight Center (MSFC) in support of ISS experiments. • Attended International Work Group Meeting on MSL-EML at DLR, Cologne, Germany, May 24-25, 2016. • Attended TT SOL-EML Meeting, Schnellenberg, Germany, May 26-27. • Developed a numerical model for multiphase droplet oscillation, which is being applied to help analyze the oscillation behavior of the Cu75Co25 batch 1 sample. • Image processing code for the analysis of MSL-EML data is being adjusted (expect to process the data downloaded from ISS this fall). Students Supported: Jie Zhao, PhD expected 2017. MSL-EML Status: Batch 1.2a: ~100 cycles, Nov/Dec 2015 VIT106a (#13), NiTa (#15), FeCo (#6, re-test) Batch 1.2b: ~250 cycles, Mar-Jun 2016 FeCrNi (#9, 34 cycles performed), FeCo (#7), FeCo (#6), Cu75Co25 (#3 & #43), Cu89Co11 (#4, 2 cycles performed), Zr (#18, 26 cycles performed), VIT106 (#21, 18 cycles performed), LEK (#10) Batch 1.2c: ~230 cycles, Ongoing MC2 (#11), CMSX-10 (#23), Zr (#18), NdFeB (#22), Ni60Al40 (#1 & #42), TiAlV (#12) Batch 2: Samples – sample holder integration completed ; Parameter development in progress. Batch 3: ESA/DLR agreement in place to start procurement in 2016.
Bibliography Type:	Description: (Last Updated: 03/30/2026)
Books/Book Chapters	Allanore A, Bartlett L, Wang C, Zhang L, Lee J, Eds. "EPD Congress 2016 : proceedings of symposia sponsored by the Extraction & Processing Division (EPD) of the Minerals, Metals & Materials Society (TMS) held during TMS 2016 145th Annual Meeting & Exhibition, February 14-18 Downtown Nashville, Tennessee." Ed. A. Allanore, L. Bartlett, C. Wang, L. Zhang, and J. Lee, Hoboken, New Jersey : John Wiley & Sons, Inc., 2016., Feb-2016